$$|A\rangle + (-|A\rangle) = 0.$$

6. Given any ket |*A*> and any complex number *z*, you can multiply them to get a new ket. Also, multiplication by a scalar is linear:

$$|zA\rangle = z|A\rangle = |B\rangle.$$

7. The distributive property holds:

$$z\{|A\rangle + |B\rangle\} = z|A\rangle + z|B\rangle$$
$$\{z + w\}|A\rangle = z|A\rangle + w|A\rangle.$$

Axioms 6 and 7 taken together are often called *linearity*.

Ordinary 3-vectors would satisfy these axioms except for one thing: Axiom 6 allows a vector to be multiplied by any complex number. Ordinary 3-vectors can be multiplied by real numbers (positive, negative, or zero) but multiplication by complex numbers is not defined. One can think of 3-vectors as forming a real vector space, and kets as forming a complex vector space. Our definition of ket-vectors is fairly abstract. As we will see, there are various concrete ways to represent ket-vectors as well.

### 1.9.2 Functions and Column Vectors

Let's look at some concrete examples of complex vector spaces. First of all, consider the set of continuous complex-valued functions of a variable *x*. Call the functions *A*(*x*). You can add any two such functions and multiply them by complex numbers. You can check that they satisfy all seven axioms. This example should make it obvious that we are talking about something much more general than three-dimensional arrows.